

- Q.19 Compare and contrast linear & rotary actuators. (CO4)
- Q.20 Explain the working of Bourdon tube with the help of suitable diagram. (CO2)
- Q.21 What is basic principle of Inductive transducer? Explain? (CO1)
- Q.22 Discuss one advantage and one challenge of using smart actuators in industrial automation. (CO5)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Explain the construction and working of electromagnetic flow meter. List its applications? (CO3)
- Q.24 Explain the working principle of a DC motor with an appropriate diagram? Also, list its advantages, disadvantages, and applications when used as an actuator. (CO3)
- Q.25 Critically analyze how smart sensors enhance system performance while also introducing potential security risks. Support your analysis with real-world examples. (CO5)

No. of Printed Pages : 4
Roll No.

222843/212843

4th Semester/ Automation & Robotics Subject : Sensors & Actuators

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 What is the basis for classifying sensors as active or passive? (CO1)
- a) Shape and size
 - b) Need for external power supply
 - c) Sensitivity
 - d) Range of measurement
- Q.2 What can a capacitive proximity sensor detect that an inductive sensor typically cannot? (CO2)
- a) Only metallic objects
 - b) Only liquid levels
 - c) Both metallic & non-metallic objects
 - d) Only magnetic fields

- Q.3 How does a Bourdon tube pressure sensor work? (CO2)
- By sensing light intensity
 - By detecting deformation of a curved tube
 - By measuring magnetic fields
 - By measuring capacitance changes
- Q.4 What is a key property of a thermistor? It/Its (CO3)
- Resistance varies with temperature
 - Detects magnetic fields
 - Generates voltage from pressure
 - Measures light intensity
- Q.5 Which material is commonly used in RTDs (Resistance Temperature Detectors)? (CO3)
- Glass
 - Rubber
 - Platinum
 - Plastic
- Q.6 What is the primary power source for a pneumatic actuator? (CO4)
- Magnetic fields
 - Hydraulic fluid
 - Electrical current
 - Compressed air

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 A standard pneumatic actuator uses both electrical and pneumatic control. [True/False] (CO4)

- Q.8 A rotary actuator produces linear displacement as its main output. [True/False] (CO4)
- Q.9 Hydraulic actuators use compressed air as their main power source. [True/False] (CO4)
- Q.10 What is the primary function of a bio-sensor? [True/False] (CO5)
- Q.11 Give one advantage of micro sensors? (CO5)
- Q.12 Piezoelectric transducers generate voltage when subjected to _____. (CO1)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Compare resistance and capacitive transducers. Give suitable examples. (CO1)
- Q.14 Explain the working principle of a thermocouple. How is it used in temperature measurement? (CO3)
- Q.15 Describe the working of a hydraulic actuator. Provide one application in industrial systems. (CO4)
- Q.16 What are smart sensors? Explain their working principle with example. (CO5)
- Q.17 What is the role of solid-state switches in electrical actuating systems. Give one example. (CO4)
- Q.18 Describe the working of a bio-sensor. Provide one application in a manufacturing context. (CO5)